

The Wage and Mobility Effects of Non-Compete Bans: Evidence from Minnesota

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1 Abstract

Non-compete agreements (NCAs) have faced increased scrutiny from federal and state governments in recent years due to their restrictions on worker mobility. In July 2023, Minnesota became just the fourth state to enact a comprehensive NCA ban. Using data from the Quarterly Workforce Indicators from 2021 to 2025, I construct a difference-in-differences framework to determine the causal effect of this policy change on key labor market indicators, including earnings and labor mobility. I use both the state of Wisconsin and a synthetic control as counterfactuals. Unlike similar research on other states, my findings suggest that Minnesota's NCA ban had marginally significant negative effects on overall earnings, along with slightly positive effects on labor mobility in the short run. This analysis provides one of the first windows into the labor effects of Minnesota's NCA ban while adding insight into how the empirical findings connect to existing models of wage-setting.

2 Introduction: Research Question & Motivation

Non-compete agreements (NCAs) are contracts between employers and employees that bar employees from working for an employer's competitor or starting a competing business in their employer's line of work for a fixed period of time after leaving a company. Over the past several years, these contracts have become subject to increasing political scrutiny from federal and state governments. Federal regulators considered a nationwide NCA ban in 2024 (a proposal that was later dropped) (Fiddler & Drab, 2025). Although only a few states have full bans on NCAs, most states have implemented "partial" bans, often restricted to covering employees within certain wage thresholds or within certain industry. In Oregon, for example, NCAs are only enforceable among employees earning over \$100,533 (Economic Innovation Group, 2026). In Hawaii, NCAs are banned entirely for employees in the technology sector, an industry where NCAs are particularly prevalent (Starr 2021). In the first four months of 2026, 13 partial NCA bans have been proposed across all U.S. states. NCA regulation has accelerated to the point where just 10 states have no restrictions on NCAs at all (Economic Innovation Group, 2026).

In July 2023, Minnesota joined California, Oklahoma, and North Dakota as one of the few states to implement a comprehensive ban on NCAs within state lines, spanning across industries and income levels. Specifically, the statute defines "covenants not to compete" (NCAs) as agreements between employers and employees barring the employee from (1) working from another employer for a specified time period, (2) working in a specified geographic area, and (3) working for another employer in a similar capacity to one's current employer. All such contracts were rendered void and unenforceable in nearly every case (Minnesota Revisor's Office, n.d.).

Because of the ban's recency, research on its effects is largely absent from economics literature. More broadly, research on the labor market effects of any statewide (full or partial) NCA bans is limited. **Thus, this project seeks to understand the short-term effects of Minnesota's NCA ban on labor market indicators.** Specifically, I will

investigate the ban’s effects on overall earnings, earnings for all hires, labor mobility, and replacement hiring rates. To accomplish this, I construct a difference-in-difference model that estimates the ban’s causal effect on the aforementioned labor market indicators and incorporate heterogeneity based on age, education, firm size, and industry. My analysis is twofold; I start by using the state of Wisconsin as the control group (making the parallel trends assumption), then transition into using a synthetic control to construct a more reliable counterfactual. Lastly, I discuss existing literature on wage-setting and monopsony to assess how my empirical results align (or do not align) with proposed theoretical models in NCA research.

3 Literature Review & Contribution

A primary inspiration for this project is a paper by Balsubramanian et al. (2022) studying the labor market effects of Hawaii’s NCA ban within the technology industry. The authors used data from the Quarterly Workforce Indicators (QWI) on earnings and labor mobility to estimate the causal effect of the ban. A selected portion of their results are displayed in Figure 1.

The authors found that the ban had a statistically significant short-run effect on average monthly earnings, which was especially pronounced among hires. Based on the full-sample results, average monthly earnings rose by 0.7% overall and by 4.2% for hires, suggesting further economic significance. Although not shown in this draft, the authors found that the ban’s effect on labor mobility was also significant and positive.

These findings are mostly consistent with other (scarce) empirical literature on the economic effects of NCA bans. Lipsitz and Starr (2021), for example, find that Oregon’s 2008 ban on NCAs for hourly-wage workers increased their wages by 2-3%, along with their labor mobility. Additionally, Young (2024) finds similar positive mobility effects resulting from an NCA ban for low-wage workers in Austria, but ambiguous effects on earnings.

Table 2*QWI Wage and Mobility Analysis from the Hawaii Natural Experiment—Triple Difference Results*

	Log Overall Average Monthly Earnings				Log Hires Average Monthly Earnings			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: QWI Wage Variables								
Post $\times$ HI $\times$ Tech	0.00964*** (0.00282)		0.00712** (0.00270)		0.0441*** (0.00457)		0.0424*** (0.00361)	
SR_Post $\times$ HI $\times$ Tech		0.0121*** (0.00276)		0.0100*** (0.00238)		0.0558*** (0.00479)		0.0548*** (0.00352)
LR_Post $\times$ HI $\times$ Tech		0.00451 (0.00653)		0.00104 (0.00546)		0.0198*** (0.00625)		0.0166*** (0.00593)
Observations	166,529	166,529	208,728	208,728	164,140	164,140	205,828	205,828
R-squared	0.992	0.992	0.993	0.993	0.975	0.975	0.975	0.975

(continued)

Figure 1: Primary regression results from Balsubramanian et al. (2022).

Note that columns 3-4 and 7-8 specify results for the full sample, while 1-2 and 5-6 specify comparisons to other industries within the 2-digit QWI technology sector code. Control group is the 40 states closest to Hawaii in NCA enforceability.

Another paper by Starr (2021) provides extensive descriptive information on the incidence of NCA use across demographic groups. Specifically, NCAs are most common among employees aged 22-40 and having earned a Bachelor's Degree or higher. Their variation with firm size exhibits more polarity, as NCAs are most prevalent among firms with fewer than 25 employees and those with more than 5,000 employees. Finally, NCAs are most prevalent in information, mining/extraction, and science/technology industries. See selected visualizations below (Figure 2 & Figure 3).

Lastly, I also plan on exploiting existing wage-setting and monopsony literature that model the wage and mobility effects of NCAs. David Card's (2022) survey paper will be a strong starting point for familiarizing myself with recent research in this area. There also appears to be an abundance of literature applying these concepts to NCA debates, specifically. Gao (2020), for example, estimates that rising NCA coverage explains about 50% of aggregate labor mobility, while Gottfries and Jarosch (2023) construct a model that projects large wage increases resulting from NCA bans. An innovation and knowledge diffusion-based model by Liu (2023) also suggests that NCAs are associated with decreased worker mobility.

Table 4: Sample Means by Noncompete Use

Variable	Overall	Bound by Noncompete?			Δ Relative to “No” Group	
		No	Maybe	Yes	Maybe	Yes
1(Employer Size < 25)	0.23	0.25	0.23	0.15	-0.02 ⁺	-0.10**
1(Employer Size 25–100)	0.16	0.16	0.16	0.15	-0.00	-0.00
1(Employer Size 101–250)	0.09	0.1	0.09	0.1	-0.01	0.00
1(Employer Size 251–500)	0.07	0.08	0.06	0.09	-0.01	0.02 ⁺
1(Employer Size 501–1,000)	0.07	0.07	0.07	0.07	0.01	0.00
1(Employer Size 1,001–2,500)	0.07	0.06	0.07	0.07	0.01	0.01
1(Employer Size 2,501–5,000)	0.07	0.06	0.08	0.08	0.02*	0.02*
1(Employer Size > 5,000)	0.24	0.23	0.24	0.29	0.01	0.06**
1(Multi-Unit Employer)	0.63	0.61	0.62	0.73	0.00	0.12**

Figure 2: Incidence of NCAs by firm size, Starr (2019)

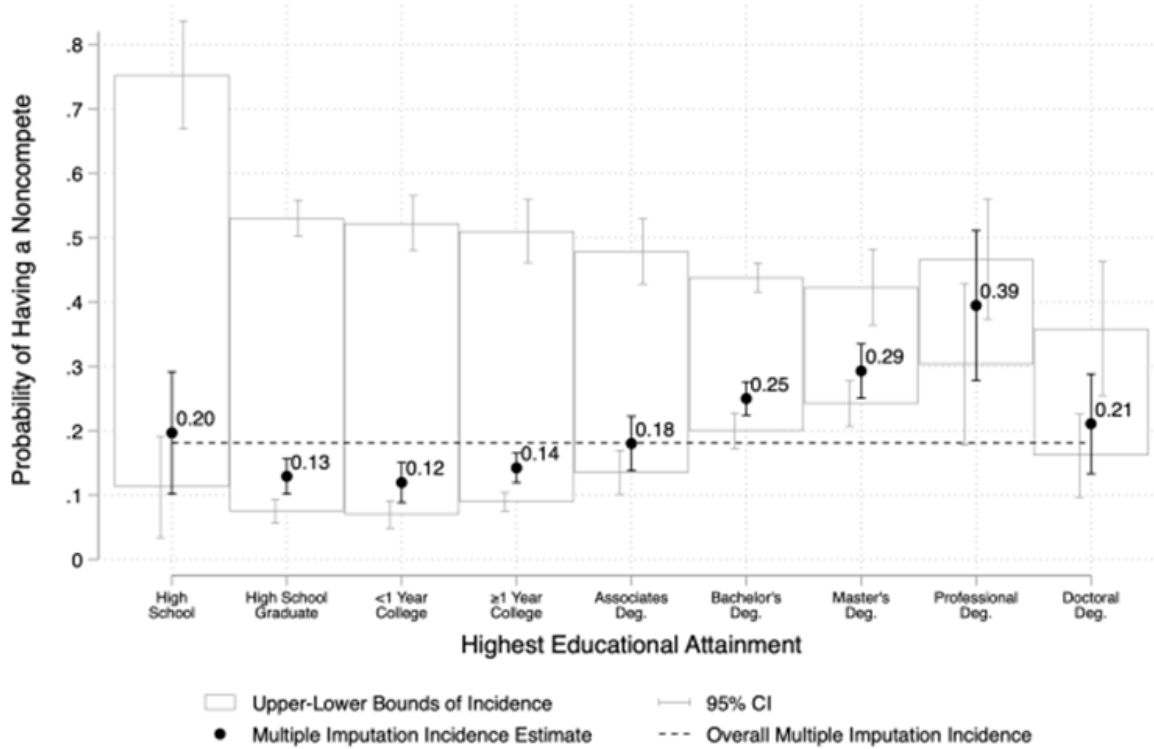


Figure 3: Incidence of NCAs by educational attainment, Starr (2019)

My project contributes to the literature in three ways. First, as mentioned previously, it provides one of the first empirical analyses on the economic effects of Minnesota’s comprehensive NCA ban. Second, it extends prior studies by incorporating heterogeneity based on age, education, firm size, and industry. Lastly, my project specifies its empirical results in the context of wider theoretical literature and models on wage-setting, monopsony, and NCAs. To this end, I will seek to assess how my results do (or do not) align with proposed models in current literature.

4 Data & Model Specification

Similar to Balsubramanian et al. (2022), I use data from the U.S. Census Bureau’s Quarterly Workforce Indicators (QWI), a quarterly panel dataset on key labor market indicators, such as earnings, separations, employment, etc. QWI data is extensive, accessible, and enhances comparability between my results from Minnesota and Balsubramanian et al. (2022). Along with labor market indicators, the QWI enables users to stratify the data based on NAICS industry codes, educational attainment, age, firm size, and more. I used an API to extract data from Q1 2021 through Q1 2025 (latest available QWI data as of April 2026). For my primary regressions (non-heterogeneity), my units are time-state-industry pairs (ex: industry 11 for Minnesota in Q1 2021), where industries are NAICS 2-digit codes to add additional observations.

The QWI data, however, has several limitations, largely related to missing data. For all states, the QWI data does not extend past Q1 2025, which limits my post-treatment window (sample size). In my sample period (Q1 2021 through Q1 2025), data from Alaska and Michigan are missing entirely. Data from Missouri is missing starting in Q3 2024, Massachusetts is missing starting in Q4 2024, and Florida is missing in Q1 2025. Other states are missing data for particular industries. The District of Columbia is missing data from industries 61 (Educational Services) and 92 (Public Administration) in 2024. Maryland

is missing data from industry 61 from Q1-Q3 2021. Lastly, Delaware is missing industry 21 (Mining, Quarrying, and Oil and Gas Extraction) in various pre-treatment periods. For all states, firm size-related heterogeneity data is not available after Q4 2024. The following sections will describe how missing data issues were addressed in the analysis.

My preliminary (full-sample) model is specified below.

$$Outcome_{tsi} = \beta_0 + \beta_1 post * MN + \alpha_t + \theta_s + \gamma_i + \epsilon_{tsi}$$

To estimate the wage effects of Minnesota’s NCA ban, following Balsubramanian et. al. (2022), I analyze two QWI wage variables: logged average monthly earnings for *all* workers (QWI: $\ln(EarnS)$) and logged average monthly earnings for all *hires* ($\ln(EarnHirAS)$). For these variables, a positive coefficient estimate can be expected if Minnesota’s NCA ban improves the wage bargaining position of workers. Under this framework, the effect should be particularly concentrated among hires, since hires usually bargain for their wage before being hired, while current employees bargain for their wage less frequently and face greater institutional constraints on pay raises. Because I am using time-state-industry units and earnings vary significantly among NAICS 2-digit industries, I log the wage variables to interpret coefficients as percent changes, which is more meaningful when industries have different wage levels.

Because NCAs constrain the employment options of workers who leave their company, a comprehensive NCA ban is also expected to have positive effects on labor mobility, since workers are no longer contractually restricted from moving to competitors. To capture these labor mobility changes, I calculate labor mobility as the ratio of separations to total employment (QWI: $Sep/EmpTotal$) within a time-state-industry pair. Furthermore, rising labor mobility also suggests an uptick in replacement hiring for firms to replace departing workers. Thus, the last outcome variable I analyze is the replacement hiring rate for each time-state-industry pair (QWI: $HirAEndReplr$).

Lastly, I include fixed effects for time, state, and industry to capture unobserved, time-invariant changes unrelated to Minnesota’s NCA ban. All standard errors are clustered at

the state-industry level to limit serial correlation. The interaction term estimates the change in the outcome variable, in Minnesota, following the enactment of the NCA ban in Q3 2023. This is the coefficient of interest for the analysis.

5 WI Control Results

The state of Wisconsin has several characteristics that make it (intuitively) suitable as a control group for this analysis. Of course, Wisconsin has close geographic proximity to Minnesota, and thus likely shares a similar demographic and industrial structure. Moreover, Wisconsin has no meaningful NCA regulation or partial ban (Economic Innovation Group, 2026), increasing the reliability of the counterfactual and making the state a meaningful unit for comparison to Minnesota.

Underlying all difference in difference models is the parallel trends assumption: if not for the treatment (here, Minnesota’s NCA ban), both the treated (Minnesota) and control (Wisconsin) units would have evolved similarly over time. I test this assumption through the following model, where t is a time trend standardized such that $t = 0$ in Q1 2021 (the beginning of the sample period).

$$Outcome_{tsi} = \beta_0 + \beta_1 t * MN + \alpha_t + \theta_s + \gamma_i + \epsilon_{tsi}$$

The coefficient estimates for all outcome variables are not statistically significant. This indicates that there is not enough evidence to conclude that pre-treatment differential trends exist such that the parallel trends assumption is violated. Importantly, however, this test only identifies linear differential trends and does not offer evidence for nonlinear differential pre-treatment trends. Subsequent analysis will include a test capable of capturing non-linear pre-treatment trends.

Table 1: Parallel Trends Test (Linear Diffenretial Pre-Treatment Trends)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hiring Rate
$MN * t$	0.0027 (0.0020)	0.0055 (0.0040)	0.0001 (0.0006)	0.0001 (0.0002)
p-value	0.171	0.172	0.884	0.610
R^2	0.959	0.935	0.779	0.948
N	400	400	400	400

Operating under the parallel trends assumption, I estimate the causal effect of Minnesota’s NCA ban on the aforementioned labor market indicators. The coefficient on $MN * Post$ represents the additional change in Minnesota compared to Wisconsin for the corresponding outcome variable, following Minnesota’s NCA ban.

Table 2: WI Control Results (Full Data)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hire Rate
$MN * Post$	0.0118 (0.0106)	0.0128 (0.0183)	0.0056 (0.0044)	0.0020 (0.0023)
p-value	0.273	0.486	0.211	0.384
R^2	0.959	0.928	0.766	0.935
N	680	680	680	680

I estimate a 1.18% increase in overall earnings, 1.28% increase in earnings for all hires, 0.56 percentage point increase in labor mobility, and 0.2 percentage point increase in replacement hiring rate following Minnesota’s NCA ban. Although none of the estimates are statistically significant, the positive sign of the observed effects align with theoretical expectations, and are potentially economically significant. This differs slightly from Balsubramanian et. al. (2022), who observe statistically significant effects on earnings and labor mobility among tech workers resulting from Hawaii’s 2015 NCA ban for tech workers. A possible explanation for this deviation is the use of different control groups. Balsubramanian et al. (2022) use the 40

most similar states to Hawaii (equally weighted) in NCA enforceability, using an index from Starr (2019), while I restrict my control group to the state of Wisconsin, where I believe the parallel trends assumption is more plausible, although not perfect. Moreover, Hawaii’s NCA ban applied specifically to the technology industry, where NCAs are more common.

Importantly, the relative frequency of NCA use varies substantially across the U.S. Labor Force. As mentioned earlier, an analysis by Starr et. al. (2021) finds that college graduates and early-to-mid career adults (age 22-44) are much more likely to be bound by NCAs compared to the average worker. Additionally, firms with less than 50 employees, as well as firms working in information, mining/extraction, and science/technology, are more likely to require their employees to sign NCAs. Thus, I also analyze whether the effect of the NCA ban is concentrated among the following subsets of the population: (1) college graduates, (2) workers aged 22-44, (3) employees at firms of less than 50 employees, (4) employees in information, mining/extraction, or the science/technology industry.

Table 3: WI Control Results (College Graduates)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hire Rate
$MN * Post$	0.0066 (0.0135)	0.0183 (0.0246)	0.0050 (0.0034)	0.0013 (0.0015)
p-value	0.627	0.459	0.144	0.362
R^2	0.945	0.903	0.770	0.937
N	680	680	680	680

Table 4: WI Control Results (Ages 22-44)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hire Rate
$MN * Post$	0.0077 (0.0076)	0.0132 (0.0089)	0.0083*** (0.0030)	0.0025 (0.0017)
p-value	0.310	0.141	0.006	0.130
R^2	0.900	0.856	0.707	0.839
N	2,720	2,720	2,720	2,720

Table 5: WI Control Results (Firms with under 50 Employees)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hire Rate
$MN * Post$	0.031** (0.0122)	0.0627*** (0.0202)	0.0048 (0.0046)	0.0014 (0.0014)
p-value	0.013	0.003	0.304	0.310
R^2	0.931	0.902	0.600	0.855
N	1,222	1,216	1,216	1,216

Table 6: WI Control Results (Information, Mining/Extraction, Prof./Scientific/Technical)

	(1) Log Avg Monthly Earnings	(2) Log Avg Monthly Earnings, Hires	(3) Labor Mobility	(4) Replacement Hire Rate
$MN * Post$	0.0559 (0.0586)	0.0612 (0.0522)	0.0135 (0.0095)	0.0001 (0.0021)
p-value	0.928	0.294	0.217	0.951
R^2	0.607	0.397	0.543	0.906
N	102	102	102	102

The results for college graduates are only notable in that the difference in the ban's marginal effect between earnings for hires and overall earnings is much more notable (about 1.2 percentage points). This indicates that Minnesota's NCA ban had a more rapid effect on earnings for newly hired college graduates in their wage negotiations, compared to non-hires who bargain for their wage less frequently. A similar wider gap also exists among

young to mid-career employees. Furthermore, Minnesota’s NCA ban also appears to have a statistically significant effect on labor mobility, which increases by 0.8 percentage points. While the 22 to 44 age range is generously wide, both the age-related mobility increase and earnings rise for college graduates suggest that the ban did materially impact the labor decisions and bargaining power of these employees. Alternatively, the rise in mobility may be attributable to younger employees being more mobile on average.

Nevertheless, the most interesting effect of the ban appears to be concentrated among small firms (less than 50 employees). Here, overall wages rise by 3.1%, including 6.3% among hires following the ban (both statistically significant). It is commonplace at smaller firms (especially startups) to use NCAs since retaining their intellectual capital is crucial to the firm’s short and long-term outlooks. These estimates indicate that, following the NCA ban, startup employers offer higher wages to keep valuable employees and/or such employees leverage the newfound opportunity of leaving for a competitor to bargain for a higher wage. No significant impact is observed for labor mobility, suggesting that small firm responded to the ban by paying higher wages rather than risking losing intellectual capital (employees).

Lastly, the industry-specific results also warrant short discussion. There is certainly economic significance to the results; overall earnings increase by 5.6%, hires’ earnings increase by 6.1%, and (most notably) labor mobility rises by 1.4 percentage points. Again, these results are consistent with theory that banning NCAs or reducing their enforceability leads to higher wages and labor mobility from increased employee bargaining power. Importantly, because of the relatively small sample size (just 102 observations among the 3 specific industries), it becomes challenging to detect statistical significance. It is therefore important to consider the non-trivial error in the estimators when interpreting these results.

6 Synthetic Control

6.1 Motivation

However, using the state of Wisconsin as the control unit for this analysis comes with several drawbacks. These include (1) nonlinear differential pre-treatment trends, (2) sensitivity to idiosyncratic post-treatment shocks, and (3) an unreliable counterfactual. I will describe each of these factors and how they motivate the use of a synthetic control (Abadie, Diamond, and Hainmueller, 2010) for this analysis.

The prior parallel trends test (Section 5) was designed to detect the presence of linear differential trends, not nonlinear differential trends. Although a test for nonlinear differential trends is forthcoming for this analysis (will be in the final version), nonlinear differential trends can also be detected graphically. The below plots provide examples of nonlinear differential trends in the QWI data between the two states among a selected industry and outcome variable. Here, the plots oscillate significantly. They also frequently intersect and structurally diverge from each other, all of which point to the presence of nonlinear differential trends. This motivates using a synthetic control for a robustness check, since the difference-in-difference framework relies on a satisfied parallel trends assumption.

A synthetic control mitigates this issue by creating a "synthetic Minnesota" for each 2-digit NAICS industry using a weighted average of other U.S. states (donor pool). The synthetic control (state) chooses its weights to best mimic selected pre-treatment characteristics of Minnesota and thereby, a more reliable post-treatment counterfactual. Thus, a synthetic Minnesota should exhibit little to no structural divergence from the state of Minnesota in each outcome variable and industry, resulting in a more plausible parallel trends assumption and replicated trajectory.

Another deficit of the single-state synthetic control is the prospect of state-specific idiosyncratic shocks, which can distort parallel trends or lead to an unreliable counterfactual. For example, if a large employer leaves the state of Wisconsin during the post-treatment

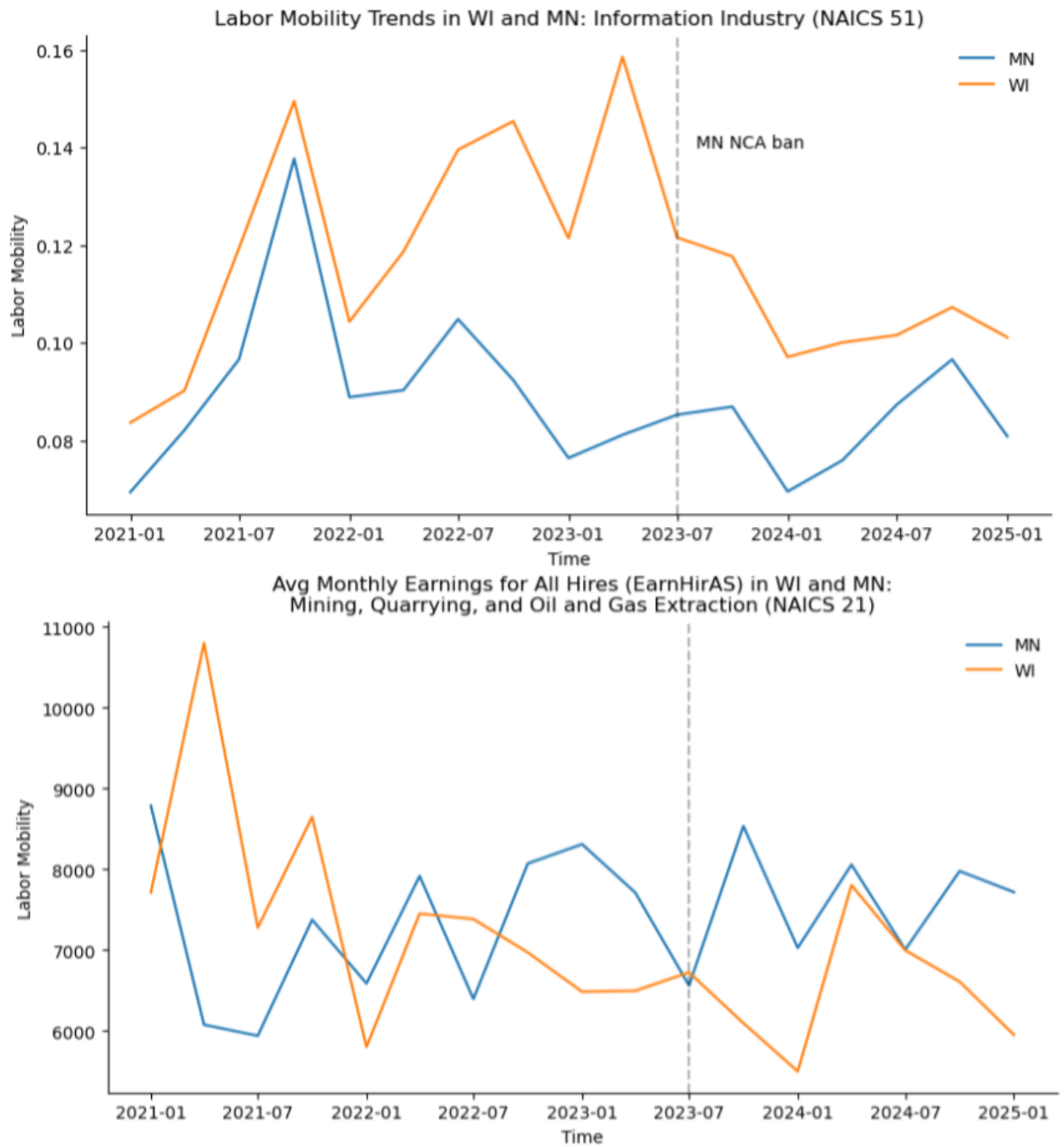


Figure 4: Selected Examples of Nonlinear Differential Trends between MN & WI

period, an event unrelated to Minnesota’s NCA ban, one would expect a decline in earnings and increased labor mobility to follow in Wisconsin. Thus, such an idiosyncratic shock would distort the difference in difference estimate for the effect of Minnesota’s NCA ban on earnings and labor mobility. A synthetic control accounts for this challenge by assigning fractional weights to several different states in the donor pool. Thus, the impact idiosyncratic shock in one particular state will be significantly reduced, improving the robustness of the synthetic control compared to a single-state control.

Finally, the synthetic control is advantageous since it produces a more reliable counterfactual than a single-state control. One of the foremost challenges in difference in difference case studies is projecting a counterfactual: what the outcomes for the treated state would have been if the unit had been left untreated. Even if all the requisite data is available, a single-state control still (at the very least) has an uncertain ability to reproduce a the treated state’s counterfactual (Abadie, Diamond, and Hainmueller, 2010). Again, because a synthetic control is designed to reproduce a synthetic Minnesota that closely tracks Minnesota’s pre-treatment outcomes, using a synthetic Minnesota becomes a far more reliable counterfactual than a single-state control, thereby producing less biased difference in difference estimators.

6.2 Data and Synthetic Control Structure

Having motivated the use of a synthetic control, I offer details on the structure of the synthetic control for this analysis. As explained above, I construct a synthetic Minnesota using a weighted average of other U.S. states (the donor pool). Specifically, the resulting weights are created such that they closely resemble Minnesota’s values for pre-treatment ”predictor” variables. Generally, predictors consist of variables that affect a given outcome variable (such as earnings), along with lags. Because my analysis works with state-industry pairs, I select predictors correlated with outcome variables within a specific industry. My chosen predictors are state unemployment rate, state per capita income, industry shares

within a state-time pair (calculated as a fraction of total employment), college graduate share within a state-time-industry pair, and small firm shares (no more than 50 employees) within a state-time-industry pair. I also include the appropriate outcome variable (such as *EarnS*) in the corresponding synthetic control. All predictors (except unemployment rate) exhibit a statistically significant relationship with each outcome variable (see Table 7).

Table 7: Synthetic Control Predictors' Correlation with Outcome Variables

	(1)	(2)	(3)	(4)
	Log Avg Monthly	Log Avg Monthly	Labor Mobility	Replacement Hire
Predictor	Earnings	Earnings, Hires	Rate	Rate
ln(Unemployment Rate)	0.0512***	0.0355***	0.0001	0.0014
ln(Per Capita Income)	0.7101***	0.6610***	-0.0426***	-0.0390***
ln(Industry Share)	-0.0966***	-0.1190***	0.0129***	0.0124***
ln(College Graduate Share)	0.6793***	0.5323***	-0.0966***	-0.0549***
ln(Small Firm Share)	-0.3175***	-0.2845***	0.0312***	0.0072***

***: $p < 0.01$

Like my outcome variables, I collect data for industry shares, college graduate shares, and small firm shares from the QWI. Each share is calculated as a fraction of employment (QWI: *EmpTotal*) within a state-time-industry pair. For example, the industry share for NAICS industry 11 (Agriculture, Forestry, Fishing and Hunting) in Alabama for Q1 2021 is calculated as the proportion of total employment in that industry to total employment in Alabama in Q1 2021. Unemployment rate data is from the U.S. Bureau of Labor Statistics, and per capita income data is from the U.S. Bureau of Economic Analysis.

Lastly, I consider the donor pool of states eligible to be weighed in constructing a synthetic Minnesota. As mentioned earlier, some states are missing QWI data for the relevant outcome variables. For states where 2 or more quarters of data are missing (Alaska, Michigan, Missouri, Massachusetts, District of Columbia), I drop these states from the donor pool entirely. For Florida, which is missing all data for Q1 2025 (the last period of my sample), I estimate an $AR(4)$ to forecast Florida's predictor values for Q1 2025. This involves using

the previous four periods (Q1-Q4 2024) of data in Florida to predict the values of predictor variables for Q1 2025. I drop all observations in Maryland for industry 61 due to missing data, as well as industry 21 in Delaware. Furthermore, the QWI only provides firm size heterogeneity until Q4 2024, not in Q1 2025. For simplicity, I carry over the Q4 2024 values for firm size shares into Q1 2025 in constructing the synthetic control, since firm size shares remain relatively constant over time. Finally, I exclude all states with a full NCA ban (California, North Dakota, and Oklahoma) like post-treatment Minnesota from the donor pool, since such states would be ineffective in estimating the counterfactual case where Minnesota does not ban NCAs.

6.3 Results

With a synthetic Minnesota created, I estimate the same model as in the prior section (Wisconsin Control). The results from these regressions can be viewed below.

$$Outcome_{tsi} = \beta_0 + \beta_1 t * MN + \alpha_t + \theta_s + \gamma_i + \epsilon_{tsi}$$

Table 8: Regression Results: Synthetic Control

	(1)	(2)	(3)	(4)
	Log Avg Monthly Earnings	Log Avg Monthly Earnings, Hires	Labor Mobility	Replacement Hire Rate
<i>MN * Post</i>	-0.0172* (0.0102)	0.0001 (0.0147)	0.0019 (0.0031)	-0.0018 (0.0020)
p-value	0.099	0.995	0.532	0.371
R^2	0.971	0.953	0.859	0.962
N	680	680	680	680

*: $p < 0.10$

These results are strikingly different from the Wisconsin control and tell a much different

story. With the synthetic control, average monthly earnings decrease by 1.7% following the ban, while earnings among hires are equivalent. Labor mobility rises by 0.2 percentage points, and replacement hiring rate falls by about 0.2 percentage points. The difference in the earnings indicators, in particular, is very notable. Rather than reflecting an improved bargaining position by employees no longer bound by NCAs, these results suggest that mobility restrictions by NCAs may have been already been "priced in" to employees' wages. Thus, when NCAs were removed, the contracts were no longer priced into employee wages. An NCA ban could also decrease wages, since firms will face reduced incentives to invest in job training if workers have a greater capability to leave the firm. Thus, employee productivity may decline, along with wages. Moreover, this could be an example of Minnesota's NCA being offset by employers increasing usage of other mechanisms that prevent employees from leaving, such as NDAs and non-solicitation agreements. Finally, there may have been idiosyncratic shocks in Minnesota itself (not any of the control states) leading to lower wages in the state which exceed the magnitude of the NCA ban, at least in the short term.

To test robustness, I also conducted the analysis using two additional alternative control specifications. First, instead of a synthetic control with all states as potential, I estimated the difference in difference model using all states as a pooled control group. Second, I limited the potential donors in my synthetic control to solely states with no regulation or ban on NCAs in the post-treatment period (per Economic Innovation Group, 2026), which included Nebraska, Kansas, Mississippi, South Carolina, North Carolina, Ohio, West Virginia, and Wisconsin (Alaska and Michigan excluded due to data collection issues). In both of these alternative control specifications, I observed similar coefficient estimates to the above synthetic control. This reflects positively on the reliability of my synthetic control. Formal regression tables for these specifications will be included in the forthcoming final draft of the paper.

Admittedly, these results vary significantly from the (scarce) existing literature on NCA bans. Balsubramanian et al. (2022) and Lipsitz and Starr (2021) both find significant positive wage and mobility effects following partial NCA bans in Hawaii and Oregon, respectively.

Young (2024) finds ambiguous effects on earnings, but significant positive effects on mobility from a partial ban in Austria. My analysis, on the other hand, finds significant effects on earnings (at the 10% level) and moderately positive effects on mobility from Minnesota’s NCA ban. Importantly, all three aforementioned analyses concern partial NCA bans in high-incidence ”subsectors” of the labor market. Hawaii’s 2015 NCA ban only covered employees in the tech industry, Oregon’s NCA ban concerned employees working for an hourly wage (who typically earn less overall than salaried workers), and the Austrian ban only covered low-wage workers. The differences between my results and the consensus of the literature may motivate continued application of my synthetic control into areas of heterogeneity (as in the WI control), such as education, age, industry, and firm size.

A notable difficulty with my synthetic control results is its inconsistent pre-treatment fit across industries and outcome variables. While the synthetic Minnesota fits some pre-treatment industries very well, others are much further off and do not bear much resemblance to the state of Minnesota itself (see selected examples in Figure 5). These results suggest that, for some industries and/or some outcome variables, the synthetic control is unreliable, which casts doubt on the results as a whole. Even after dropping more volatile industries driving many of the inconsistencies, the difference in difference estimates remain largely unchanged. For my final draft, I first plan on addressing this difficulty by tinkering with the predictors used in constructing the synthetic control, aiming to select factors that can more closely project each outcome variable. Another potential solution may be to use the same state weights for each industry, rather than an individual set of weights for each industry-outcome variable pairing.

Lastly, I note that the QWI data exhibits a high degree of seasonality, where outcome variables’ values tend to peak consistently in certain quarters or otherwise present a seasonal trend. For example, across states and industries, average monthly earnings tend to peak in the first quarter of each year. These peaks are followed by dips, and subsequent increases in the next Q1, thus following a seasonal trend (Figure 6).

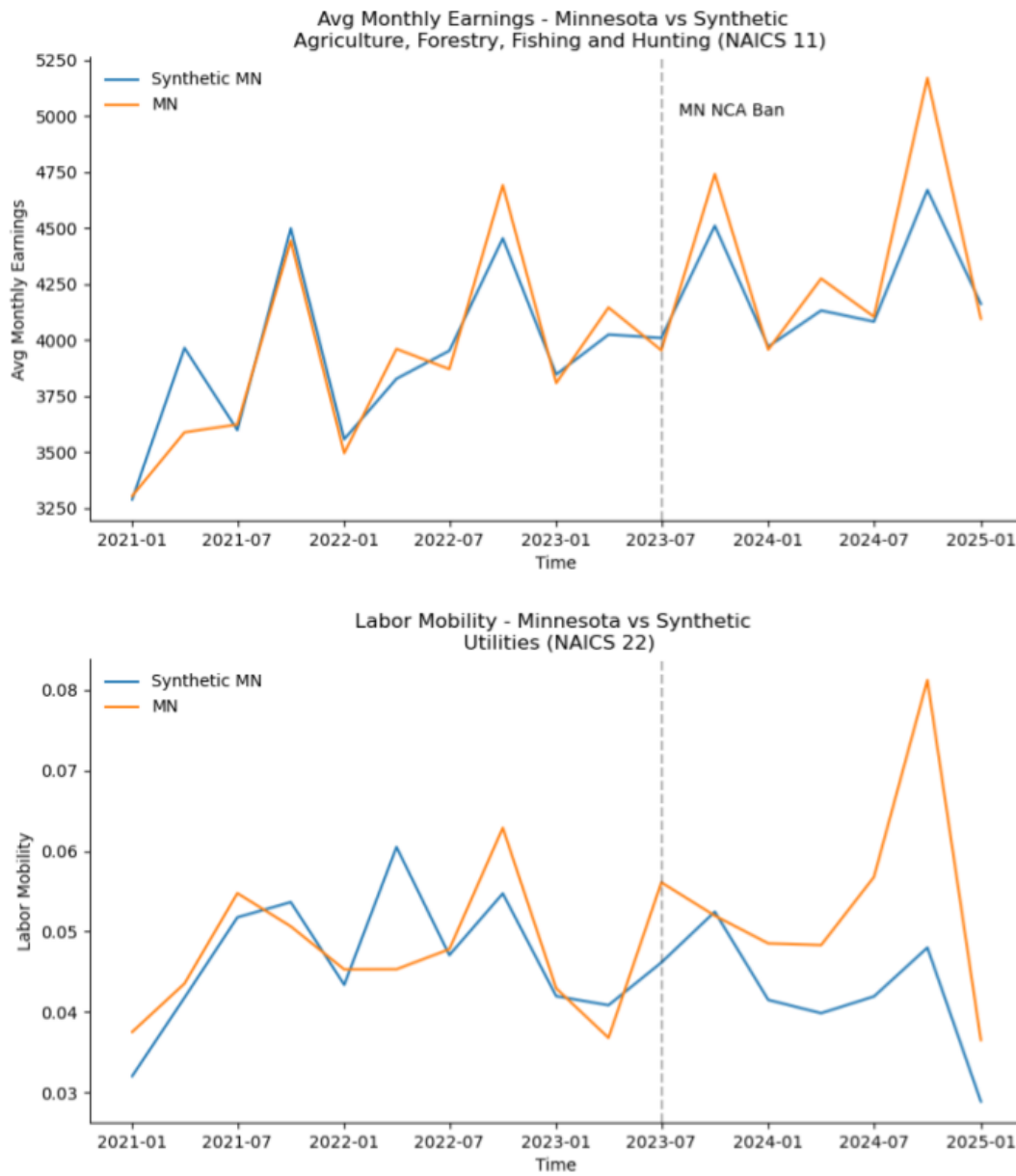


Figure 5: Examples of Pre-Treatment Synthetic Fit

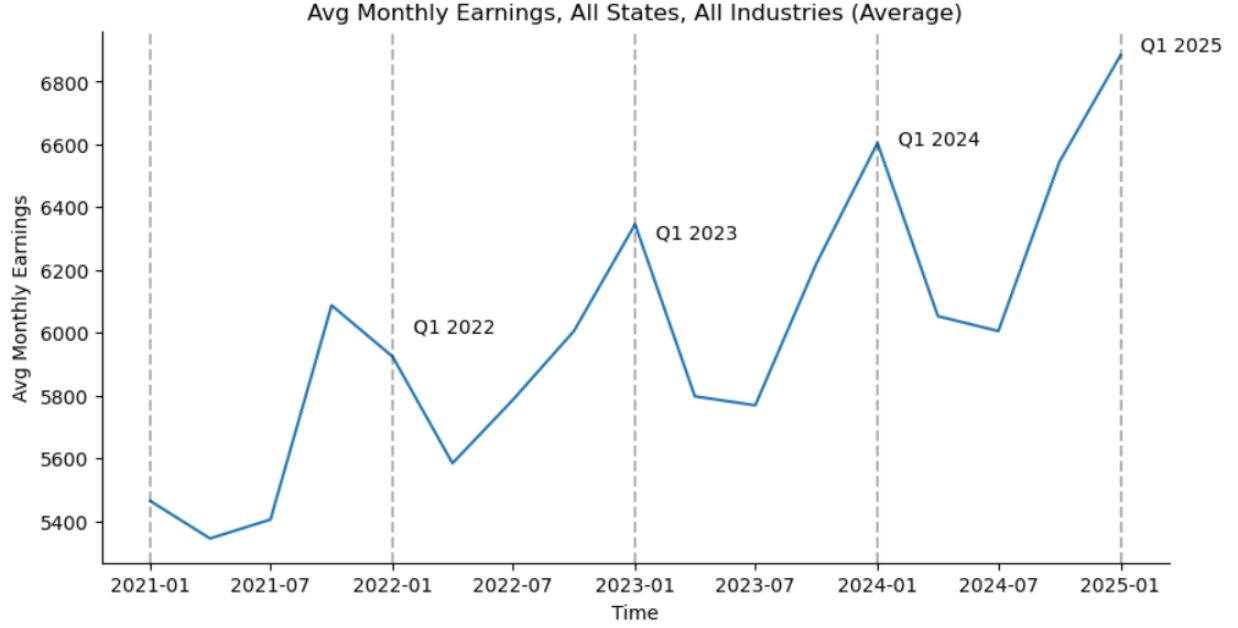


Figure 6: Example of Seasonality in Outcome Variables

To address this seasonality problem and as an added robustness check for my synthetic control, I re-define the values for my synthetic control using the variation in my predictors that is unexplained by the seasonal trend. To accomplish this, I estimate the following model:

$$Predictor_{tsi} = \alpha_0 + \alpha_1 Q_1 + \alpha_2 Q_2 + \alpha_3 Q_3 + \epsilon_{tsi}$$

Here, Q_1 , Q_2 , and Q_3 are dummy variables indicating which quarter an observation was made in. To avoid perfect multicollinearity, I omit a dummy variable for Q_4 and absorb its seasonal effect through the constant α_0 . After estimating the model, I predict the residuals for the full sample, which represent the variation in my predictors that is not explained by the seasonality (which quarter it is). I then use this variation to re-construct my synthetic control and run the difference in difference regressions. The results of these regressions are forthcoming (will be in the final draft of my paper), and I expect the removal of seasonal patterns to produce more unbiased estimators.

7 Notes for Reviewers

Currently, my results appear ambiguous, and I am not sure which results are and are not reliable in describing a causal effect of the NCA ban on labor market indicators in Minnesota. I would appreciate any suggestions in addressing the synthetic control's fitting problem (such as alternative predictors I could use), or anything else that can help explain the results I have obtained. The direction of my estimators is unusual compared to the bulk of (scarce) existing literature. This is not necessarily negative, but it is important for me to be able to defend my results well. Please also point out any motivations/explanations in my project that are difficult to follow or could otherwise use more clarity. If there are any visualizations and/or summaries I do not have included that would be helpful in developing my analysis (or would be interesting to include), please also let me know. Thank you for your time and effort.

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